
ROBOTICS & INTELLIGENT SYSTEMS

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MODULE 1

Introduction to Robotics & Robot Configurations	Introduction to robotics – Degrees of freedom – Robot types
	Manipulators- Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller
	Robot configurations-PPP, RPP, RRP, RRR
Mobile Robots	Mobile robots- wheeled, Legged robots
	Aerial robots, underwater robots, surface water robots
Characteristics, Effectors & Grippers	Dynamic characteristics of robot- speed of motion, load carrying capacity & speed of response
	Introduction to End effectors - mechanical grippers, special tools, Magnetic grippers
	Vacuum grippers, adhesive grippers, Active and Passive grippers
Robotics Ethics	Ethics in robotics - 3 laws - applications of robots

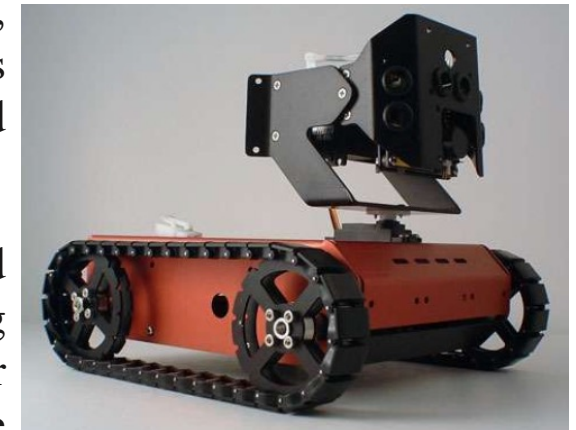
Introduction To Robotics

What Is A Robot ?

- A programmable machine that imitates the actions or appearance of an intelligent creature – usually a human
- a machine able to extract information from its environment and use knowledge about its world to act safely in a meaningful and purposeful manner (Arkin, 1998)
- Requirements:
 - Must have sensing and perception (i.e. gets information from its environment)
 - Must be able to complete different tasks (i.e. locomotion or manipulation)
 - Re-programmable for different tasks or actions
 - Functions autonomously or interacts with human beings

What Is Robotics ?

- *Robotics* is the art, knowledge base, and know-how of designing, applying, and using robots in human endeavours. Robotic systems consist of not just robots, but also other devices and systems that are used together with the robots.
- Robots may be used in manufacturing environments, in underwater and space exploration, in researching human and animal behaviour, for aiding the disabled, for transportation and delivery, for military purposes, or even for fun. In any capacity, robots can be useful but need to be programmed and controlled. Robotics is an interdisciplinary subject that benefits from mechanical engineering, electrical and electronic engineering, computer science, cognitive sciences, biology, and many other disciplines.



Classification of Robots

- Fixed-sequence robot: A device that performs the successive stages of a task according to a predetermined, unchanging method that is hard to modify.
- Playback robot: A human operator performs a task manually by leading the robot, which records the motions for later playback. The robot repeats the same motions according to the recorded information.
- Numerical-Control Robot: The operator supplies the robot with a movement program rather than teaching it the task manually.
- Intelligent Robot: A robot with the means to understand its environment and the ability to successfully complete a task despite changes in the surrounding conditions under which it is to be performed.

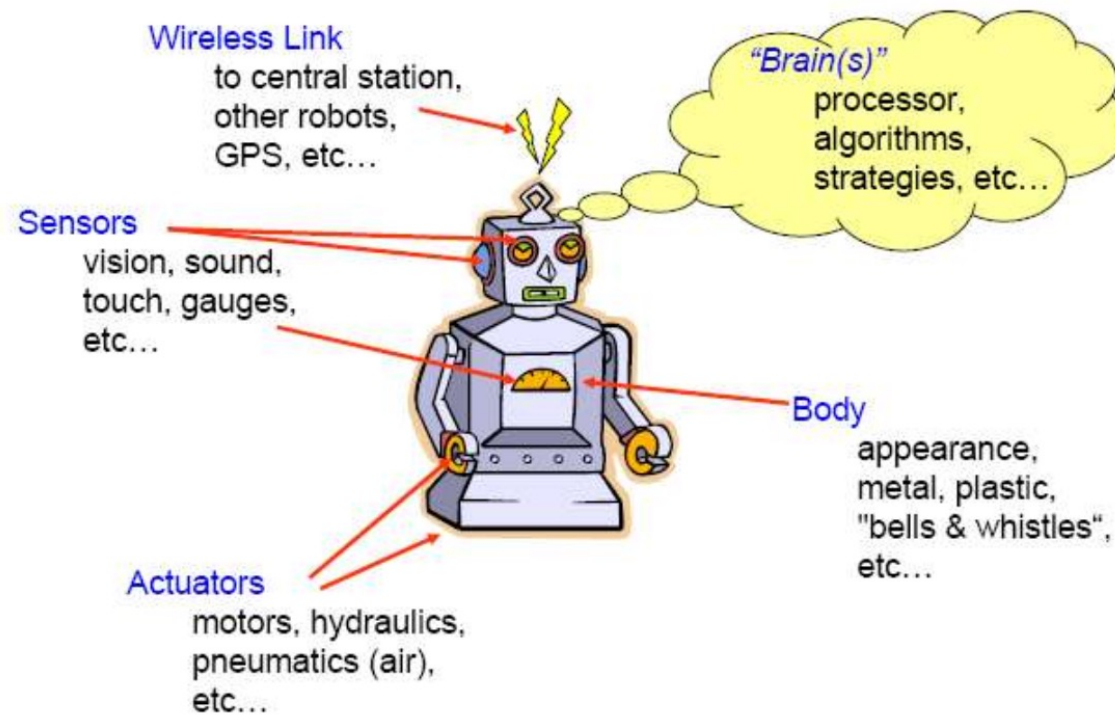
History of Robotics

Year	Events and Development
1954	First patent on manipulator by George Devol , the father of robot
1956	Joseph Engelberger started the first robotics company: Unimation
1962	General Motors used the manipulator: Unimate in die-casting application
1967	General Electrical Corporation made a 4-legged vehicle
1969	1. SAM was built by the NASA, USA 2. Shakey, an intelligent mobile robot, was built by Stanford Research Institute (SRI)
1970	1. Victor Scheinman demonstrated a manipulator known as Stanford Arm 2. Lunokhod I was built and sent to the moon by USSR 3. ODEX 1 was built by Odetics

History of Robotics (Contd.)

Year	Events and Development
1973	Richard Hohn of Cincinnati Mialcron Corporation manufactured T³ (The Tomorrow Tool) robot
1975	Raibart at CMU, USA, built a one-legged hopping machine, the first dynamically stable machine
1978	Unimation developed PUMA (Programmable Universal Machine for Assembly)
1983	Odetics introduced a unique experimental six-legged device
1986	ASV (Adaptive Suspension Vehicle) was developed at Ohio State University, USA
1997	Pathfinder and Sojourner was sent to the Mars by the NASA, USA
2000	Asimo humanoid robot was developed by Honda
2004	The surface of the Mars was explored by Spirit and Opportunity

Components of a Robot



Components of a Robot (Contd.)

- **Power Supply** - The working power to the robot is provided by batteries, hydraulic, solar power, or pneumatic power sources.
- **Actuators** - Actuators are the energy conversion device used inside a robot. The major function of actuators is to convert energy into movement.
- **Electric motors (DC/AC)**- Motors are electromechanical component used for converting electrical energy into its equivalent mechanical energy. In robots, motors are used for providing rotational movement.

Components of a Robot (Contd.)

- **Sensors** - Sensors provide real time information on the task environment. Robots are equipped with tactile sensor it imitates the mechanical properties of touch receptors of human fingerprints and a vision sensor is used for computing the depth in the environment.
- **Controller** - Controller is a part of robot that coordinates all motion of the mechanical system. It also receives an input from immediate environment through various sensors. The heart of robot's controller is a microprocessor linked with the input/output and monitoring device. The command issued by the controller activates the motion control mechanism, consisting of various controller, actuators and amplifier.

Components of a Robot (Contd.)

- **Processor.** The processor is the brain of the robot. It calculates the motions of the robot's joints based on the programs it runs, determines how much and how fast each joint must move to achieve the desired location and speeds, and oversees the coordinated actions of the controller and the sensors.
- The processor is generally a computer, which works like all other computers but is dedicated to this purpose. It requires an operating system, programs, and peripheral equipment like a monitor, and it has the same limitations and capabilities. In some systems, the controller and the processor are integrated together into one unit.
- In others, they are separate units. And in some, although the controller is provided by the manufacturer, the processor is not; the manufacturer expects the user to provide their own processor.

Components of a Robot (Contd.)

- **Software.** Three groups of software programs are used in a robot.
 1. The operating system operates the processor.
 2. Robotic software calculates the necessary motions of each joint based on the kinematic equations of the robot. This information is sent to the controller. This software may be at many different levels, from machine language to sophisticated languages used by modern robots.
 3. A collection of application- oriented routines and programs are developed in order to use the robot or its peripherals for specific tasks, such as assembly, machine loading, material handling, and so on. This includes additional vision routines when the robot is equipped with a vision system.

Types Of Robot

According to Structural Capability of Robot

- **Mobile robot:** A mobile robot is an automatic machine that is capable of locomotion. .
 - Example: spying robot. Mobile robots have the capability to move around in their environment and are not fixed to one physical location.
 - Mobile robots can be "autonomous" (AMR - autonomous mobile robot) which means they are capable of navigating an uncontrolled environment without the need for physical or electro-mechanical guidance devices.
 - Alternatively, mobile robots can rely on guidance devices that allow them to travel a pre-defined navigation route in relatively controlled space (AGV – autonomous guided vehicle).
 - By contrast, industrial robots are usually more-or-less stationary, consisting of a jointed arm (multi-linked manipulator) and gripper assembly (or end effector), attached to a fixed surface
- **Fixed Robot:** Most industrial robots are fixed with the base, but the arms are moving.

According to Control

- To perform as per the program instructions, the joint movements an industrial robot must accurately be controlled. Micro-processor-based controllers are used to control the robots. Different types of control that are being used in robotics are given as follows.
- **Limited Sequence Control:**
 - It is an elementary control type. It is used for simple motion cycles, such as pick-and- place operations. It is implemented by fixing limits or mechanical stops for each joint and sequencing the movement of joints to accomplish operation.
 - Feedback loops may be used to inform the controller that the action has been performed, so that the program can move to the next step. Precision of such control system is less. It is generally used in pneumatically driven robots.

According to Control (Contd)

○ Playback with Point-to-Point Control

- Playback control uses a controller with memory to record motion sequences in a work cycle, as well as associated locations and other parameters, and then plays back the work cycle during program execution.
- Point-to-point control means individual robot positions are recorded in the memory. These positions include both mechanical stops for each joint, and the set of values that represent locations in the range of each joint. Feedback control is used to confirm that the individual joints achieve the specified locations in the program.

According to Control (Cont.)

- Play Back with Continuous path control:
 - Continuous path control refers to a control system capable of continuous simultaneous control of two or more axes.
 - The following advantages are noted with this type of playback control: greater storage capacity—the number of locations that can be stored is greater than in point-to-point; and interpolation calculations may be used, especially linear and circular interpolations.

According to Control (Cont.)

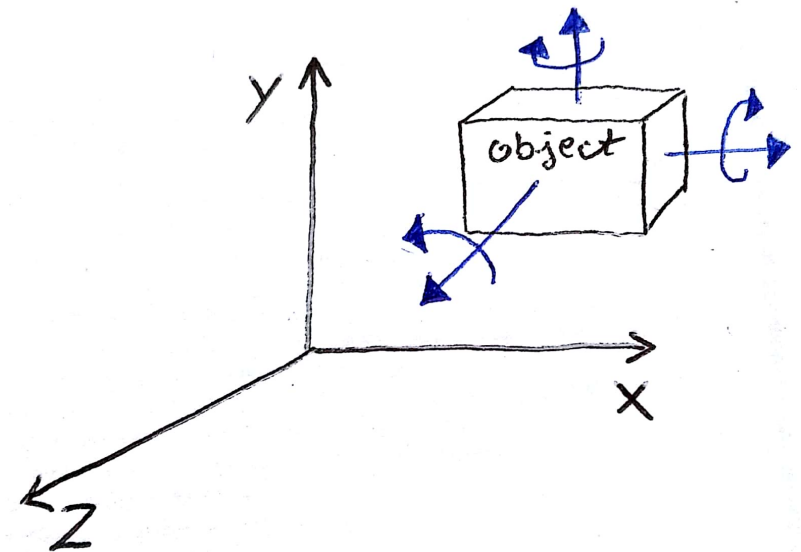
○ Intelligent Control

- An intelligent robot exhibits behaviour that makes it seem to be intelligent.
- For example, it may have capacity to interact with its ambient surroundings; decision-making capability; ability to communicate with humans; ability to carry out computational analysis during the work cycle; and responsiveness to advanced sensor inputs.
- They may also possess the playback facilities. However, it requires a high level of computer control, an advanced programming language for decision-making logic and other intelligence' into the memory.

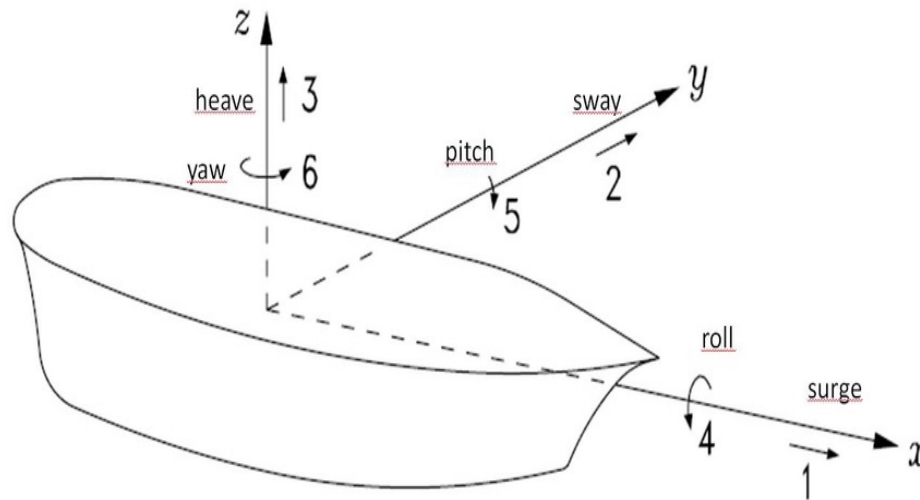
Degrees Of Freedom

Degrees of Freedom

- In mechanics, the degree of freedom (DOF) of a mechanical system is the number of independent parameters that define its configuration.
- It is the number of parameters that determine the state of a physical system and is important to the analysis of systems of bodies in mechanical engineering, aeronautical engineering, robotics, and structural engineering.
- The position and orientation of a rigid body in space is defined by three components of translation and three components of rotation, which means that it has six degrees of freedom.



Degrees of Freedom(Cont.)



○ **Example:** The motion of a ship at sea has the six degrees of freedom of a rigid body, and is described as shown

○ **Translation:**

1. Moving up and down (heaving);
2. Moving left and right (swaying);
3. Moving forward and backward (surging);

○ **Rotation:**

1. Tilts forward and backward (pitching);
2. Swivels left and right (yawing);
3. Pivots side to side (rolling).

Degree of freedom (Contd.)

- In order to locate a point in space, we need to specify three coordinates (such as x -, y -, and z -coordinates along the three Cartesian axes). Three coordinates are necessary and adequate to completely define the location of the point.
- Although different coordinate systems may be used to express this information, they are always necessary. However, neither two nor four will be possible; two is inadequate to locate a point in space, and four is illogical as there is simply too much information. For example, a device like a crane can move to any location within its workspace.
- To locate a rigid body (a three-dimensional object rather than a point) in space, we first need to specify the location of a selected point on it, and therefore we require three pieces of information. Next, we also need to specify the orientation of the object to fully specify it in space. This means that six pieces of information are needed to fully specify the location and orientation of a rigid body. By the same token, there need to be 6 degrees of freedom (DOF) available to fully place the object in space and also orientate it as desired.

Degree of freedom (Contd.)

- For this reason, robots need to have 6 DOF to be able to freely place and orientate objects within their workspace. A robot that has 6 DOF can be requested to place objects at any desired location and orientation. If a robot has fewer DOF, we cannot arbitrarily specify any location and orientation for the robot; it can only go to places and to orientations that the fewer joints allow.
- To demonstrate this, consider a robot with 3 DOF, where it can only move along the x -, y -, and z -axes. In this case, no orientation can be specified; all the robot can do is pick up a part and move it in space parallel to the reference axes. The orientation always remains the same.
- Now consider another robot with 5 DOF, capable of rotating about the three axes but only moving along the x - and y -axes. Although you may specify any orientation desired, the positioning of the part is only possible along the x - and y -, but not z -, axes. The same is true for any other robot configurations.

Degree of freedom (Contd.)

- A system with 7 DOF would not have a unique solution. This means that if a robot has 7 DOF, there are infinite ways it can position a part and orient it at the desired location. In order for the controller to know what to do, there must be an additional decision-making routine that allows it to pick only one of the infinite solutions. As an example, we may use an optimization routine to pick the fastest or the shortest path to the desired destination. Then the computer has to check all solutions to find the shortest or fastest response and perform it. Due to this additional requirement, which can take much computing power and time, no 7-DOF robot is used in industry.
- A similar issue arises when a manipulator robot is mounted on a moving base such as a mobile platform or a conveyor belt. In either case, the robot has an additional degree of freedom, which, based on the previous discussion, is impossible to control. The robot can be at a desired location and orientation from infinite distinct positions on the conveyor belt or mobile platform. However, in this case, the additional DOF are known, and there is no need to solve for them.

Degree of freedom (Contd.)

- When a robot is mounted on a conveyor belt or a track, as shown or is otherwise mobile, the location of the base of the robot relative to the belt or other reference frame is known. Since this location does not need to be defined by the controller, the remaining number of DOF is still 6 and, consequently, unique.
- As long as the location of the base of the robot on the belt or the location of the mobile platform is known (or selected by the user), there is no need to find it by solving the set of equations of robot motions; and as a result, the system can be solved.



The robot can move along a track , adding a degree of freedom to the system.

Degree of freedom (Contd.)

- **DOF- of Human Arm:**
- The human arm has three joint clusters: the shoulder, the elbow, and the wrist. The shoulder has 3 DOF, since the upper arm (humerus) can rotate in the sagittal plane (parallel to the midplane of the body), in the coronal plane (a plane from shoulder to shoulder), and about the humerus (verify this by rotating your arm about the three different axes). The elbow has only one degree of freedom; it can only flex and extend about the elbow joint.
- The wrist also has 3 DOF. It can abduct and adduct and flex and extend; and since the radius bone can roll over the ulna, it can rotate longitudinally (pronate and supinate). Consequently, the human arm has a total of 7 DOF, even if the ranges of some movements are small. Note that the end effector of the robot is never included in the DOF count. All robots have this additional capability, which may appear to be similar to a degree of freedom. However, none of the movements in the end effector are counted towards the robot's DOF.

Degree of freedom (Contd.)

- In some cases, the movements of a joint may not be fully controllable. For example, when a pneumatic actuator is used to move a joint, it can only be fully turned on or off (fully extended or fully retracted), but nothing in between. In this case, the convention is to assign only a half a degree of freedom to the joint.
- This means that the joint can only be at specified locations within its limits of movement. Another possibility for a half degree of freedom is to assign only particular values to the joint. For example, suppose that a joint is made to be only at 0, 30, 60, and 90 degrees. Then, as before, the joint is limited to only a few possibilities, and therefore it has a partial degree of freedom.

Degree of freedom (Contd.)

- Many industrial robots possess fewer than 6 DOF. Robots with 3.5, 4, and 5 DOF are in fact very common. As long as there is no need for the additional DOF, these robots perform very well. As an example, suppose that you intend to insert electronic components into a circuit board.
- The circuit board is always laid flat on a known work surface; consequently, its height (z value) relative to the base of the robot is known. Therefore, there is only a need for 2 DOF along the x - and y -axes to specify any location on the board for insertion.
- Additionally, suppose that the components are to be inserted in any direction about the vertical axis. In that case, there is a need for one degree of freedom to rotate about the vertical axis (z) in order to orient the component above the surface.

Degree of freedom (Contd.)

- Since there is also a need for half a degree of freedom to fully extend the end effector to insert the part or to fully retract it to lift the end effector before moving, only 3.5 DOF are needed: 2 to move over the board, 1 to rotate the component, and 0.5 to insert or retract. Insertion robots are very common and are extensively used in electronic industry.
- Their advantage is that they are simple to program, less expensive, smaller, and faster. Their disadvantage is that, although they may be programmed to insert components on any size board in any direction, they cannot perform other jobs.
- They are limited to what 3.5 DOF can achieve, but they can perform a variety of functions within this design limit.

Manipulators, Anatomy of Robot

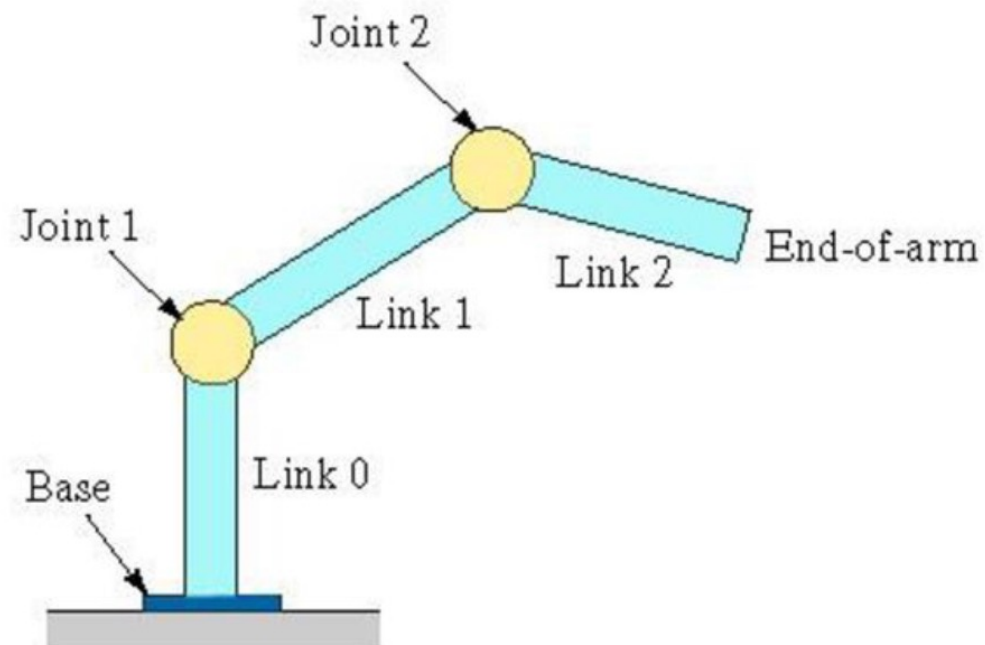
Manipulators

- The arm-like structure of an industrial robot is known as a robot manipulator. This component is responsible for completing the tasks the robot is programmed to perform. Also known as a robot arm, the manipulator mounts to the robot body and consists of multiple links and joints.
- Links are rigid components that connect different sections of the robot arm. Their rigidity provides a solid structure, but joints are necessary for flexibility. As with the human body, robotic joints move in different ways, facilitating linear, rotary, and revolutionary motions. This allows the robot to perform the desired movements within its work envelope.
- The main body of a robot consisting of the links, joints, and other structural elements, is called the manipulator. A manipulator becomes a robot when the wrist and gripper are attached, and the control system is implemented. However, in literature robots and manipulators are utilized equivalently and both refer to robots.

Robot Anatomy

- A robot as a system, consists of a manipulator or rover, a wrist, an end-effector, actuators, sensors, controllers, processors, and software. Robot Manipulators are composed of links connected by joints into a kinematic chain.
- **Link:** A robot arm or a robot link is a rigid member that may have relative motion with respect to all other links.
- **Joint:** Two links are connected by contact at a joint where their relative motion can be expressed by a single coordinate. Joints are typically revolute (rotary) or prismatic (translatory).

Robot Anatomy



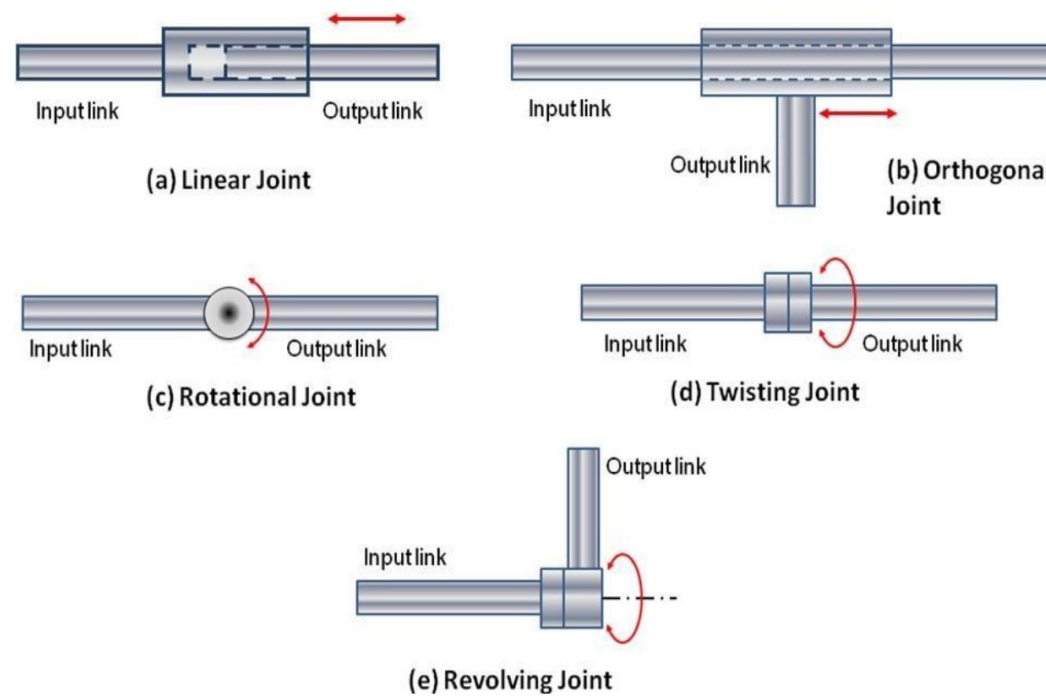
Joints & Links

- The manipulator of an industrial robot consists of a series of joints and links. Robot anatomy deals with the study of different joints and links and other aspects of the manipulator's physical construction. A robotic joint provides relative motion between two links of the robot.
- Each joint, or axis, provides a certain degree-of-freedom (dof) of motion. In most of the cases, only one degree-of- freedom is associated with each joint. Therefore, the robot's complexity can be classified according to the total number of degrees-of-freedom they possess.
- Each joint is connected to two links, an input link and an output link. Joint provides controlled relative movement between the input link and output link. A robotic link is the rigid component of the robot manipulator. Most of the robots are mounted upon a stationary base, such as the floor.

Joints & Links (Cont.)

- The robotic base and its connection to the first joint are termed as link-0. The first joint in the sequence is joint-1. Link-0 is the input link for joint-1, while the output link from joint-1 is link-1 which leads to joint-2. Thus link 1 is, simultaneously, the output link for joint-1 and the input link for joint-2. This joint-link-numbering scheme is further followed for all joints and links in the robotic systems.

Types of Joints



Types of Joints (Cont.)

- **Linear joint (type L joint) (Prismatic Joint)**
 - The relative movement between the input link and the output link is a translational sliding motion, with the axes of the two links being parallel.
- **Orthogonal joint (type U joint)**
 - This is also a translational sliding motion, but the input and output links are perpendicular to each other during the movement.
- **Rotational joint (type R joint)**
 - This type provides rotational relative motion, with the axis of rotation perpendicular to the axes of the input and output links.

Types of Joints (Cont.)

- **Twisting joint (type T joint)**
 - This joint also involves rotary motion, but the axis of rotation is parallel to the axes of the two links.
- **Revolving joint (type V-joint, V from the “v” in revolving)**
 - In this type, axis of input link is parallel to the axis of rotation of the joint. However the axis of the output link is perpendicular to the axis of rotation.

Robot Wrist

- The joints in the kinematic chain of a robot between the forearm and end effector are referred to as the wrist. It is common to design manipulators with spherical Wrists .
- The spherical wrist greatly simplifies the kinematic analysis effectively. Thus, the arm will have 6 DoF (3 for positioning and 3 for orientation). We may design a wrist having one, two, or three DOF depending on the application.

End Effector

- The end-effector is the part mounted on the last link to do the required job of the robot. The simplest end-effector is a gripper, which is usually capable of only two actions: opening and closing.
- The arm and wrist assemblies of a robot are used primarily for positioning the end-effector and any tool it may carry. It is the end-effector or tool that performs the work.
- Thus, a robot is composed of a manipulator or mainframe and a wrist plus a tool.
- The wrist and end-effector assembly are also called a hand.

Actuators

- Actuators are drivers that move the joints in robots to change their configuration.
- The actuators provide power to act on the mechanical structure against gravity, inertia, and other external forces to modify the geometric location of the robot.
- The actuators can be of electric, hydraulic, or pneumatic type and must be controllable.

Sensors

- The sensors are elements used for detecting and collecting information about internal conditions of robot and its surrounding environmental states.
- Generally, joint position, velocity, acceleration, and force are the most important information to be sensed.
- Sensors, integrated into the robot, send information about each link and joint to the control unit, and the control unit determines the configuration of the robot.

Controller

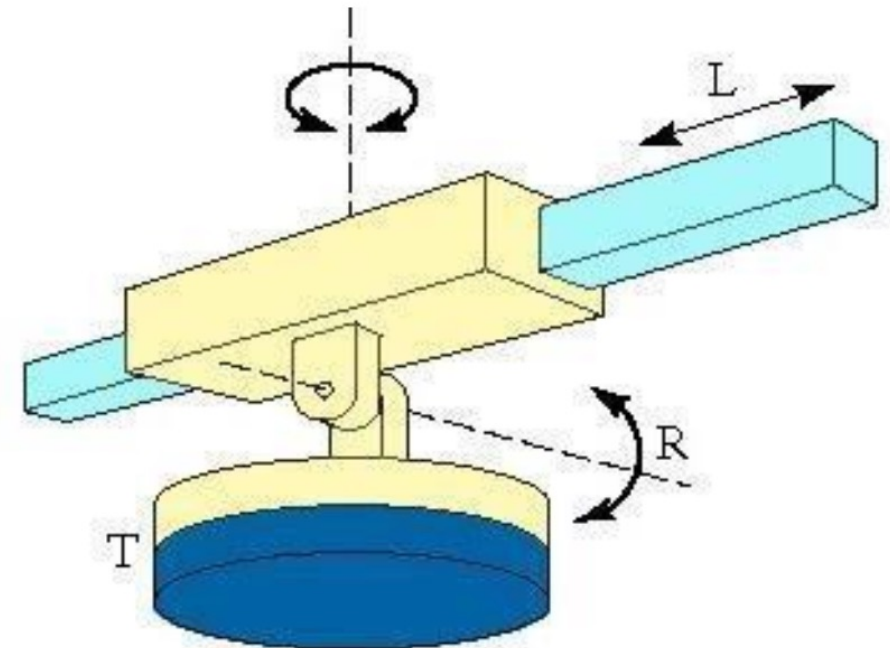
- The controller or control unit includes the processor and software. It has three roles.
 1. Information role, which consists of collecting and processing the information provided by the robot's sensors.
 2. Decision role, which consists of planning the geometric motion of the robot structure.
 3. Communication role, which consists of organizing the information between the robot and its environment.

Robotic Arm Configurations

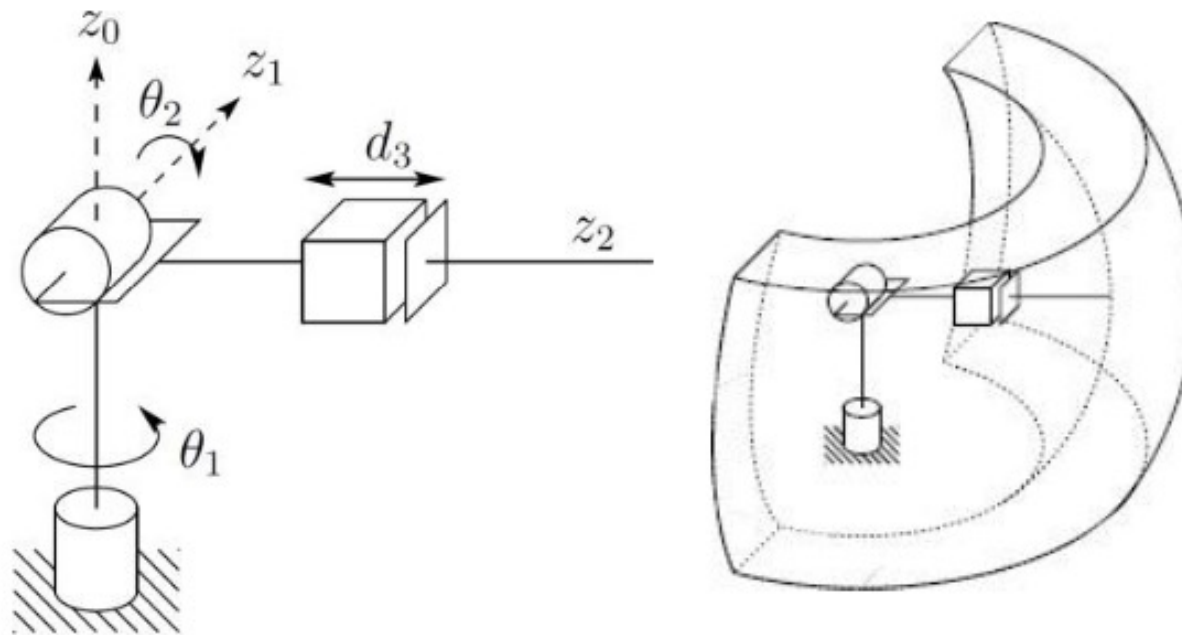
- For body-and-arm configurations, there are many different combinations possible for a three-degree-of-freedom robot manipulator, comprising any of the five joint types.
- Common body-and-arm configurations are as follows.
 - Polar coordinate arm configuration (Spherical Robots) (RRP)
 - Cylindrical coordinate arm configuration (PPR) (RPP)
 - Cartesian coordinate arm configuration (PPP)
 - Jointed arm configuration (RRR)

Polar Coordinate arm configuration (RRP)

- The polar arm configuration is shown. It consists of a prismatic joint that can be raised or lowered about a horizontal revolute joint.
- The two links are mounted on a rotating base. These various joints provide the capability of moving the arm endpoint within a partial spherical space.
- Therefore, it is called as Spherical coordinated configuration. This configuration allows manipulation of objects on the floor.



Spherical Configuration & Its workspace (RRP)

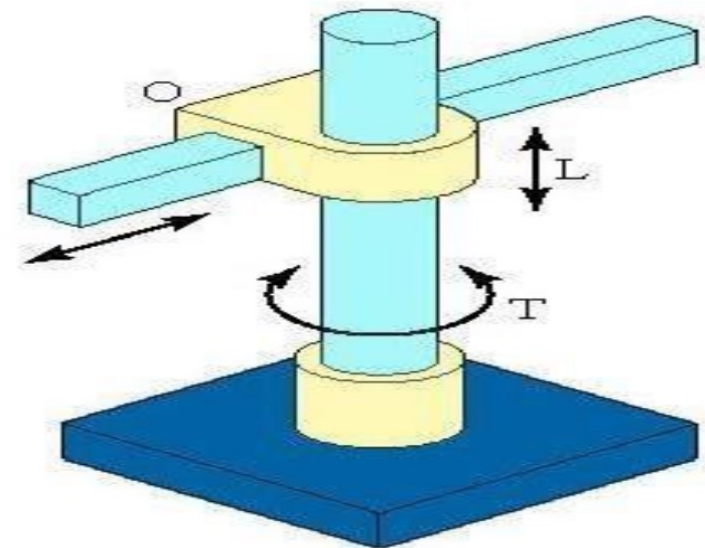


Polar Coordinate arm configuration (RRP) (CONT.)

- **Drawbacks:**
 - i. Low mechanical stiffness
 - ii. Complex construction
 - iii. Position accuracy decreases with the increasing radial stroke.
- **Applications:** Machining, spray painting
- **Example:** Unimate 2000 series, MAKER 110

Cylindrical Coordinate Arm Configuration (RPP)

- The cylindrical configuration uses two perpendicular prismatic joints and a revolute joint as shown.
- This configuration uses a vertical column and a slide that can be moved up or down along the column.
- The robot arm is attached to the slide, so that it can be moved radially with respect to column.
- By rotating the column, the robot can achieve a workspace that approximates a cylinder.
- The cylindrical configuration offers good mechanical stiffness.

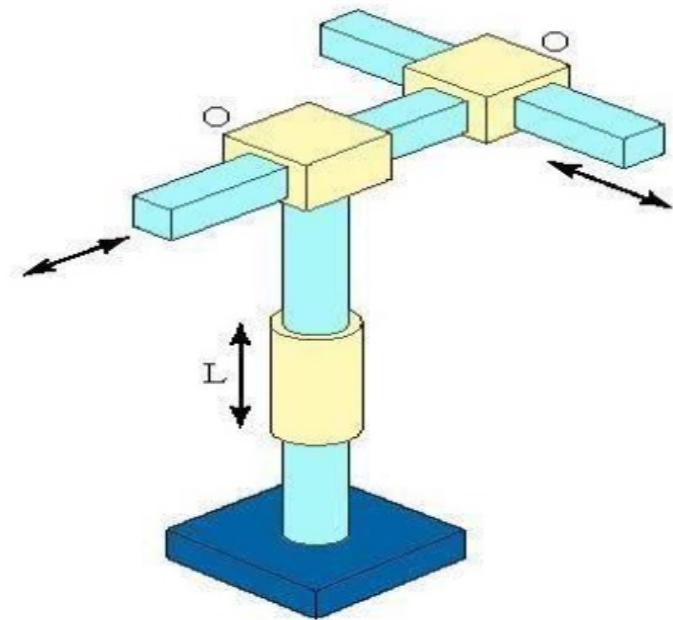


Cylindrical Coordinate Arm Configuration (RPP)

- **Drawback:** Accuracy decreases as the horizontal stroke increases.
- **Applications:** suitable to access narrow horizontal capabilities, hence used for machine
- Loading operations. **Example:** GMF model M-1A.

Cartesian Coordinate arm configuration (PPP)

- Cartesian coordinate or rectangular coordinate configuration is constructed by three perpendicular slides, giving only linear motions along the three principal axes.
- It consists of three prismatic joints.
- The endpoints of the arm can operate in a cuboidal space.
- Cartesian arm gives high precision and is easy to program.

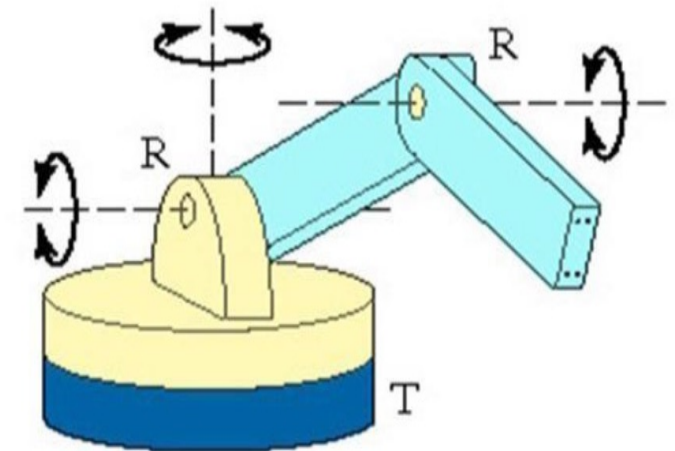


Cartesian Coordinate arm configuration (PPP)

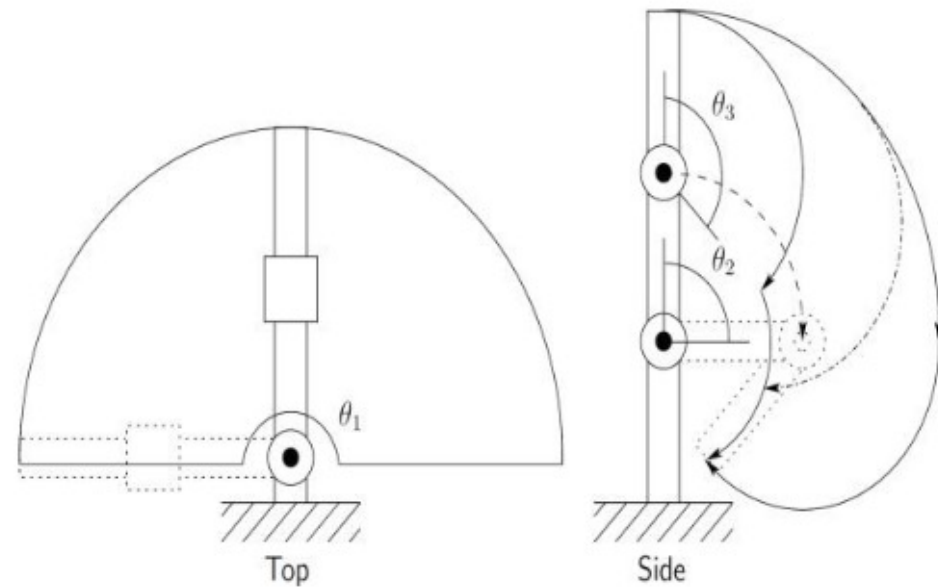
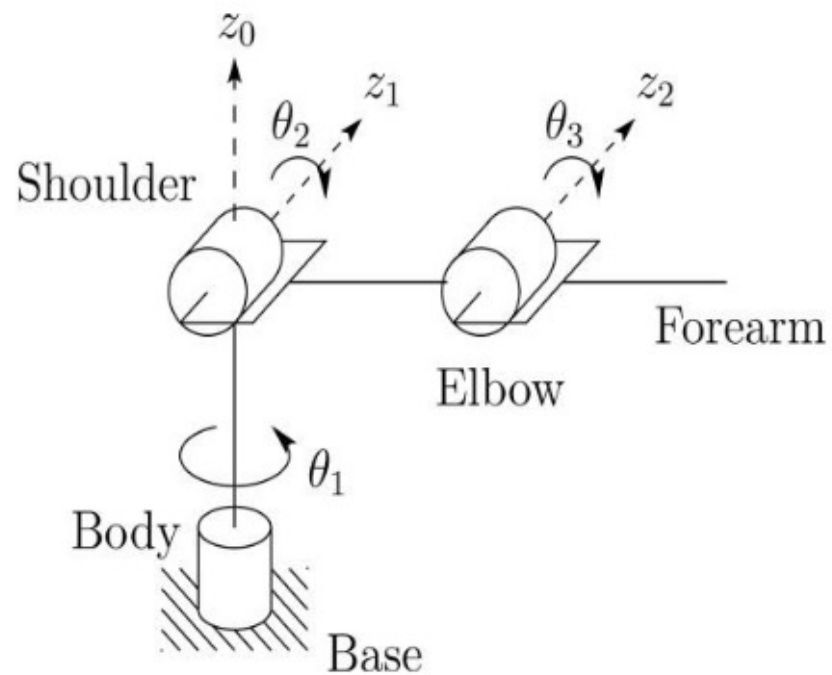
- **Drawbacks:**
 - limited manipulability
 - low dexterity (not able to move quickly and easily)
- **Applications:** use to lift and move heavy loads.
- **Example:**

Joined Arm Configuration (RRR)

- Also Known as Articulated configuration.
- Jointed arm configurations are similar to that of human arm. It consists of two straight links, corresponding to human forearm and upper arm with two rotary joint corresponding to the elbow and shoulder joints.
- These two are mounted on a vertical rotary table corresponding to human waist joint.
- The work volume is spherical.
- This structure is the most dexterous one. This configuration is very widely used.



Joined Arm Configuration (RRR)



Joined Arm Configuration (RRR)

- **Applications:** Arc welding, Spray coating.
- **Example: SCARA** robot (Selective compliance Assembly Robot Arm) Its full form is 'Selective Compliance Assembly Robot Arm'. It is similar in construction to the jointed-arm robot, except the shoulder and elbow rotational axes are vertical. It means that the arm is very rigid in the vertical direction, but compliant in the horizontal direction.
- The SCARA body-and-arm configuration typically does not use a separate wrist assembly. Its usual operative environment is for insertion-type assembly operations where wrist joints are unnecessary. The other four body-and-arm configurations more-or-less follow the wrist-joint configuration by deploying various combinations of rotary joints viz. type R and T.

SCARA Manipulator & Its Workspace

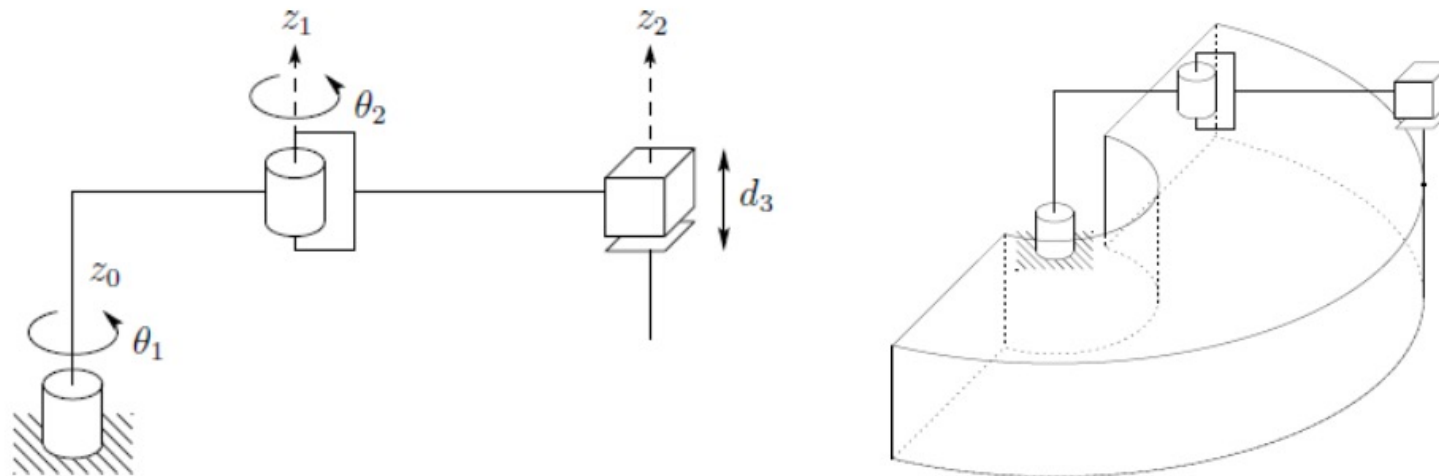


Fig: SCARA manipulator and its workspace.

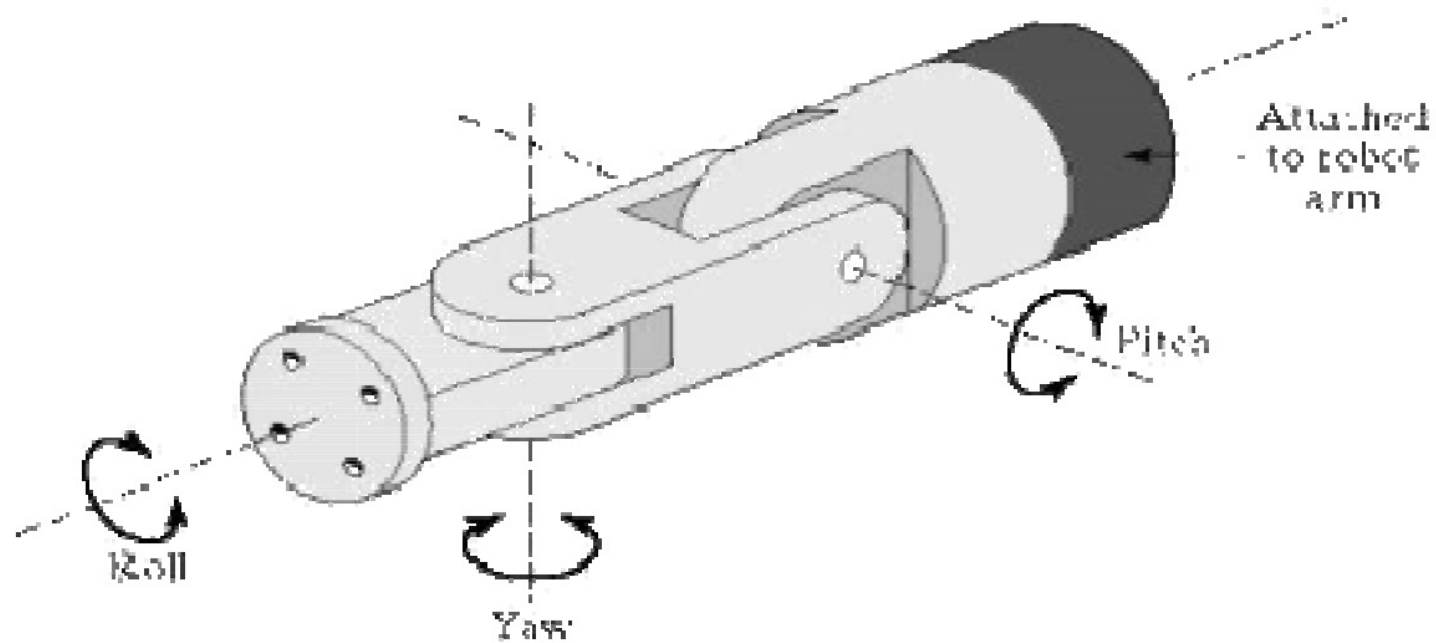
Scara Manipulator

- SCARA (Selective Compliant Articulated Robot for Assembly) is tailored for assembly operations. Although the SCARA has an RRP structure, the difference is that RRP structure has 3 axes z_0 , z_1 , z_2 mutually perpendicular but the SCARA has z_0 , z_1 , z_2 parallel.

Robot Wrist

- Wrist assembly is attached to end-of-arm. End effectors are attached to wrist assembly. Function of wrist assembly is to orient end effectors. Body-and-arm determines global position of end effector. It has three degrees of freedom:
 - Roll (R) axis – involves rotation of the wrist mechanism about the arm axis.
 - Pitch (P) axis – involves up or down rotation of the wrist.
 - Yaw (Y) axis - involves right or left rotation of the wrist.
- Robot wrist assembly consists of either two or three degrees-of-freedom. A typical three-degree-of-freedom wrist joint is depicted in Figure
- The roll joint is accomplished by use of a T joint; the pitch joint is achieved by recourse to an R joint; and the yaw joint, a right-and-left motion, is gained by deploying a second R joint. Care should be taken to avoid confusing pitch and yaw motions, as both utilize joints.

Robot Wrist



Mobile Robots

Introduction- Mobile Robots

- Mobile robots have the capability to move around in their environment and are not fixed to one physical location. Mobile robots are robots that can move from one place to another autonomously, that is, without assistance from external human operators.
- Mobile robots can be "autonomous" (AMR - autonomous mobile robot) which means they are capable of navigating an uncontrolled environment without the need for physical or electro-mechanical guidance devices.
- Alternatively, mobile robots can rely on guidance devices that allow them to travel a pre-defined navigation route in relatively controlled space (AGV - autonomous guided vehicle).

Introduction- Mobile Robots

- Unlike most industrial robots that can move only in a specific workspace, mobile robots have the special feature of moving around freely within a predefined workspace to achieve their desired goals. This mobility capability makes them suitable for a large repertory of applications in structured and unstructured environments.
- Ground mobile robots are distinguished in wheeled mobile robots (WMRs) and legged mobile robots (LMRs)
- Mobile robots also include unmanned aerial vehicles (UAVs), and autonomous underwater vehicles (AUVs).

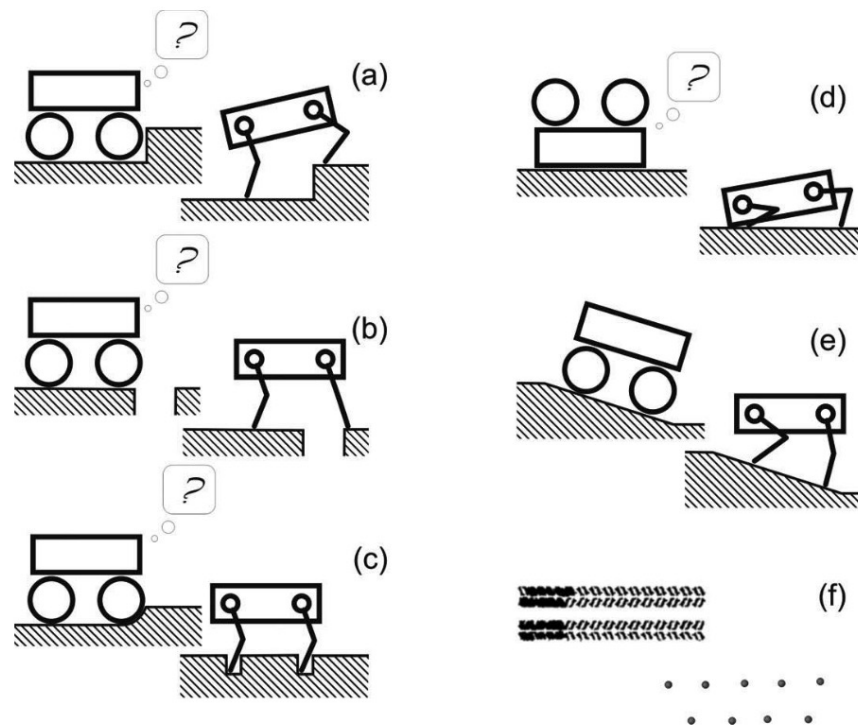
Introduction- Mobile Robots

- WMRs are very popular because they are appropriate for typical applications with relatively low mechanical complexity and energy consumption.
- Legged robots are suitable for tasks in non- standard environments, stairs, heaps of rubble, etc. Typically, systems with two, three, four, or six legs are of general interest, but many other possibilities also exist. Single-leg robots find rare applications because they can only move by hopping.
- Mobile robots also include mobile manipulators (wheeled or legged robots equipped with one or more light manipulators to perform various tasks).

Legged Robots

- Legged robots are a type of mobile robot, which use articulated limbs, such as leg mechanisms, to provide locomotion.
- They are more versatile than wheeled robots and can traverse many different terrains, though these advantages require increased complexity and power consumption.
- Legged robots often imitate legged animals, such as humans or insects, in an example of biomimicry.

Legged Mobile Robots

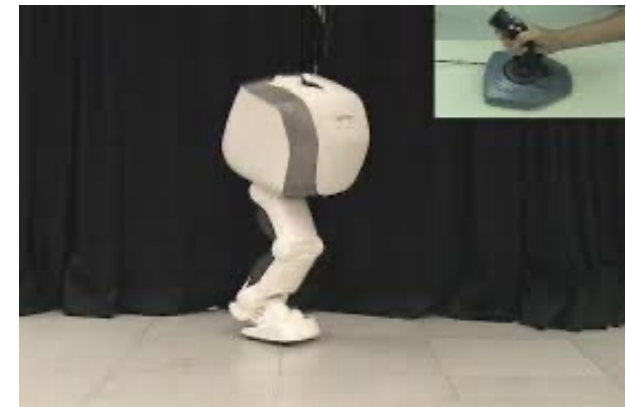


Types of Legged Robot

- One Legged Robot (Pogo stick robot)
- Two - Legged Robot (Bipedal robot)
- Four-Legged Robot (Quadrupedal robot)
- Six-Legged Robot (Hexapod)

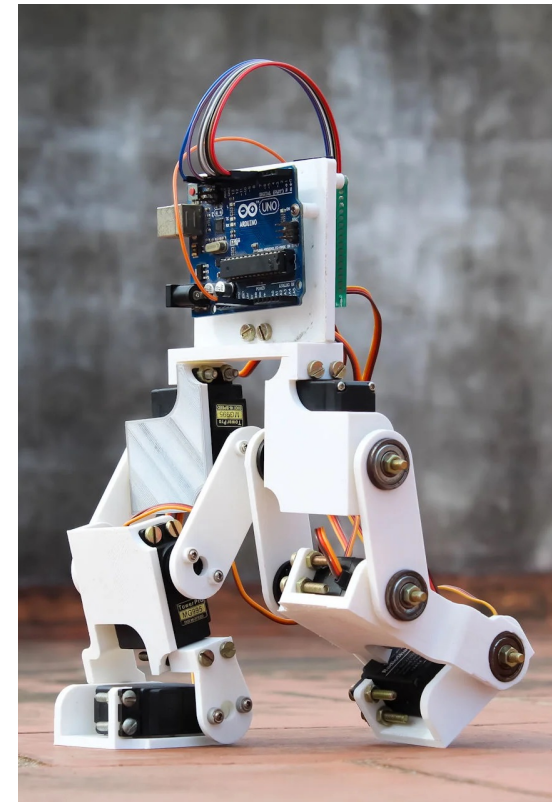
One Legged Robot

- They are stable, they can balance themselves.
- They can hop over and move in any kind of terrain as they take a running start and jump over any obstacles.
- They are very efficient, but they come with complex design & control.



Two- Legged (Bi-pedal Robot)

- Two-legged robots (Bipedal robots) can exhibit bipedal motion.
- Stability is maintained by calculating each step and moving the weight in a well-defined way, but they face two primary problems:
 - stability control that refers to the robot's balance,
 - motion control that refers to the robot's ability to move.
- Bipedal robot has 2 legs, and the classical example is the Humanoid robot.



Four-Legged (Quadrupedal)

- Quadrupedal robots are statically stable, especially when they are not in motion.
- They have 4 legs and their walking pattern similar to that of animals.
- They are well balanced in posture.
- At slow speeds, They can move either by moving one leg at a time ensuring a stable tripod or moving the alternate pair of legs to walk.



Six-Legged Robot (Hexapod)

- Six-legged robots (hexapods) offer greater stability than bipedal or quadrupedal robots. They have 6 legs.
- Their final designs mimic the mechanics of insects, and their gaits may be categorized similarly. These include the slowest wave gait, where pairs of legs move in a “wave” from the back to the front.
- Tripod gait which is a slightly faster step, where three legs move at once, the remaining three legs offer a stable tripod for the robot.

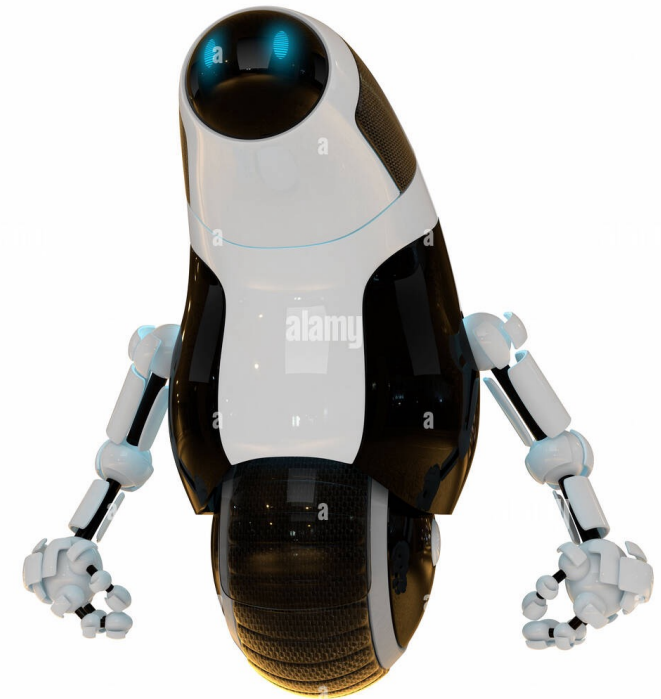


Wheeled Mobile Robot

- *Wheeled robots* are robots that navigate around the ground using motorized wheels to propel themselves. This design is simpler than using treads or legs and by using wheels they are easier to design, build, and program for movement in flat, not-so-rugged terrain. They are also more well controlled than other types of robots.
- Disadvantages of wheeled robots are that they cannot navigate well over obstacles, such as rocky terrain, sharp declines, or areas with low friction. Wheeled robots are most popular among the consumer market, their differential steering provides low cost and simplicity.
- Robots can have any number of wheels, but three wheels are sufficient for static and dynamic balance. Additional wheels can add to balance; however, additional mechanisms will be required to keep all the wheels in the ground, when the terrain is not flat.

Wheeled Mobile Robot (Contd.)

- *One Wheeled Robot:* One-wheeled robots are difficult to keep balanced due to the single point of contact with the ground, they have only one wheel, it is easier to use a spherical wheel rather than a typical disc wheel, as the robot can move in any direction along the sphere, Gyroscopes & counter torque mechanisms can keep the robot upright.



Wheeled Mobile Robot (Contd.)

- *Two Wheeled Robot* : Two-wheeled robots are simple, they have their wheels parallel to each other, and they are called dicycles, or one wheel in front of the other.
- For the robot that has the left and right wheels, it comes with at least two sensors, The tilt sensor is used to determine tilt angle & wheel encoders keep track of the position of the platform of the robot.



Aerial Mobile Robots

- Aerial mobile robots, also known as aerial drones or unmanned aerial vehicles (UAVs), are autonomous or remotely controlled aircraft designed to perform various tasks without the need for onboard human pilots.
- These robots can range from small quadcopters to larger fixed-wing aircraft and helicopters, equipped with sensors, cameras, and other payloads for specific applications.



Aerial Mobile Robots (Contd.)

- **Agriculture:** Drones equipped with cameras and sensors can monitor crops, assess plant health, detect pests and diseases, and optimize irrigation and fertilization.
- **Search and Rescue:** UAVs can be used in search and rescue missions to survey disaster areas, locate survivors, and deliver supplies to remote or inaccessible locations.
- **Mapping and Surveying:** Aerial robots equipped with LiDAR, cameras, and GPS can create detailed 3D maps, survey terrain, and monitor environmental changes.
- **Disaster Response:** Drones can assess damage after natural disasters such as earthquakes, hurricanes, and wildfires, providing real-time data to aid in disaster response and recovery efforts.
- **Construction and Mining:** Aerial robots can monitor construction sites, track progress, and conduct volumetric measurements in construction and mining operations, improving efficiency and safety.

Under-Water Robot

- An underwater robot is an autonomous underwater vehicle (AUV) that can travel underwater without requiring input from an operator. Researchers drop an AUV in the ocean pick it up at a pre-selected position.
- Two types:
 - Autonomous under water vehicle an underwater robot is an autonomous underwater vehicle (AUV) that can travel underwater without requiring input from an operator. AUVs are part of a larger group of undersea systems known as unmanned underwater vehicles.
 - Remotely operated underwater vehicles It is controlled and powered from the surface by an operator/pilot via an umbilical or using the remote control.

Surface Water Robots

- Surface Water Robots are used for the removal and collection of surface waste.
- A motor-driven collecting-arm system has been designed for collecting the wastes and redeploying it into a rectangular basket on the hull. This design provides simple and effective waste removal and accommodates large amounts of waste within a small space.
- It is used to reduce the labour intensity of the work and improve the cleaning coverage and cleaning efficiency.



Dynamic Characteristics of a Robot

- Dynamic characteristics in robotics refer to the aspects related to the motion, load carrying capacity, and response speed of a robot. These characteristics are crucial for evaluating the performance and capabilities of a robot in different applications.
- Major Dynamic Characteristics of Robot are as follows:
 1. Speed of Motion
 2. Load Carrying Capacity
 3. Speed of Response

Speed of Motion

- Speed of motion refers to how quickly a robot can move through its workspace.
- The speed of motion refers to how quickly a robot can move from one position to another. It is a crucial dynamic characteristic that impacts the efficiency and performance of robotic systems in various applications.
- The speed of motion can be influenced by factors such as the design of the robot (e.g., its actuators, motors, and mechanical components), the terrain or environment in which it operates, and the specific task it is performing.

Speed of Motion (Contd.)

- In industrial robotics, for example, high-speed motion may be desirable for increasing productivity and throughput in manufacturing processes.
- Robots used in pick-and-place operations or assembly lines often need to move swiftly between different workstations to minimize cycle times and maximize production output.
- However, in some applications, such as delicate manipulation tasks or environments with obstacles or safety constraints, lower speeds may be necessary to ensure accuracy, precision, and safety.
- In mobile robotics, such as autonomous vehicles or drones, the speed of motion must be balanced with considerations such as navigation accuracy, collision avoidance, and energy efficiency.

Load Carrying Capacity

- Load carrying capacity is the maximum payload a robot can handle while maintaining stability and performance.
- The load carrying capacity, also known as payload capacity or lifting capacity, refers to the maximum weight that a robot can carry or manipulate while maintaining stable operation.
- It is a critical dynamic characteristic, particularly in applications involving material handling, manipulation, and transportation.
- The load carrying capacity of a robot is determined by factors such as its mechanical structure (e.g., frame, joints, and actuators), the strength and durability of its components, and the stability of its control system. Robots with higher load carrying capacities are capable of handling heavier objects or payloads, expanding their range of potential applications.

Load Carrying Capacity(Contd.)

- In industrial robotics, for instance, robots with high load carrying capacities are used in tasks such as palletizing, packaging, and heavy lifting in manufacturing and logistics facilities.
- These robots can handle large and heavy objects with ease, improving productivity and reducing the need for manual labour.
- The load carrying capacity is closely related to the robot's speed of motion, as carrying heavier payloads may affect the robot's dynamic performance, including its acceleration, deceleration, and manoeuvrability.
- Therefore, it's essential to consider both factors when designing, selecting, and operating robotic systems for specific tasks and environments.

Speed of Response

- Speed of response refers to how quickly a robot can react to changes in its environment or inputs.
- The speed of response, also known as the dynamic response or responsiveness, refers to how quickly a robot can react to changes in its environment or control inputs. It is a critical dynamic characteristic that influences the agility, adaptability, and efficiency of robotic systems in dynamic and unpredictable environments.
- The speed of response depends on factors such as the robot's control system (e.g., sensors, actuators, and feedback loops), its processing capabilities (e.g., computing power and algorithms), and the latency or delay in communication between different components.

Speed of Response (Contd.)

- In applications requiring fast and precise motion, such as robotics-assisted surgery or autonomous driving, rapid response times are essential for ensuring safety, accuracy, and effectiveness. Robots must be able to react quickly to changes in their surroundings, such as avoiding obstacles, adjusting their trajectory, or responding to user commands.
- Achieving a fast speed of response often involves advanced control algorithms, real-time sensor feedback, and high-performance actuators. However, it's essential to balance speed with stability and reliability to prevent overshoot, oscillations, or other undesirable behaviors that could compromise the robot's performance and safety.

End Effectors

Introduction to End Effectors

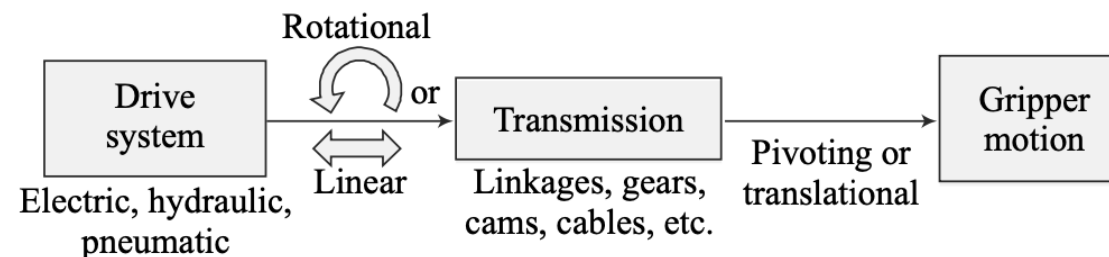
- End effectors are devices attached to the end of a robotic arm or manipulator, tasked with interacting with the environment. They play a crucial role in enabling robots to perform various tasks such as grasping, lifting, manipulating, and placing objects.
- End effectors come in a variety of types and configurations, each suited to specific applications and environments.
- In Robotics, the term end-effector is used to describe a tool that is attached to the wrist. End-effectors make a general-purpose robot to perform a particular operation.
- The End-Effectors are mainly classified into two:
 - Grippers
 - Tools

Grippers

- The function of gripper is mainly to translate some form of power into grasping action of fingers. The power input is either supplied by pneumatic, hydraulic, or electric means.

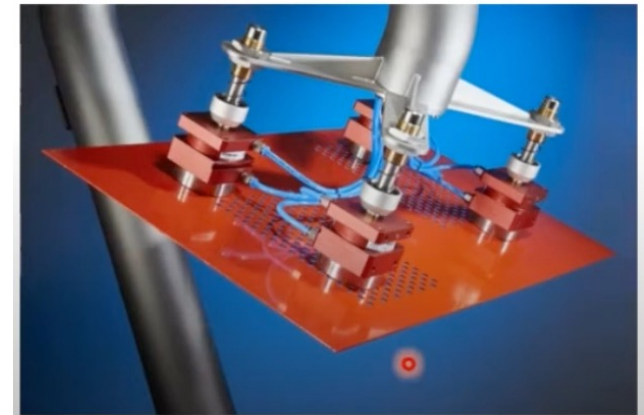
Mechanical Grippers

- Mechanical grippers are one of the most common types of end effectors, consisting of jaws or fingers that can open and close to grasp objects.
- They are versatile and can be used to handle a wide range of objects, from small parts in manufacturing to larger items in material handling.
- Mechanical grippers may come in various configurations, including parallel jaw grippers, three-fingered grippers, and adaptive grippers.



Magnetic Grippers

- Used to grip ferrous materials. Magnetic gripper uses a magnetic head to attract ferrous materials like steel plates. The magnetic head is simply constructed with a ferromagnetic core and conducting coils. Magnetic grippers are most used in a robot as end effectors for grasping the *ferrous* materials. It is another type of handling the work parts other than the mechanical grippers and vacuum grippers.
- Are of two types:
 - 1) Electromagnets
 - 2) Permanent Magnets



Magnetic Grippers (Contd.)

○ Electromagnets:

- Electromagnetic grippers include a *controller unit* and a *DC power* for handling the materials.
- This type of grippers is easy to control, and very effective in releasing the part at the end of the operation than the permanent magnets.
- If the work part gripped is to be released, the polarity level is minimized by the controller unit before the electromagnet is turned off.
- This process will certainly help in *removing the magnetism* on the work parts.
- As a result, a best way of releasing the materials is possible in this gripper.

Magnetic Grippers (Contd.)

○ Permanent Magnets

- The permanent magnets do not require any sort of external power as like the electromagnets for handling the materials.
- After this gripper grasps a work part, an additional device called as *stripper push – off pin* will be required to separate the work part from the magnet.
- This device is incorporated at the sides of the gripper. The advantage of this permanent magnet gripper is that it can be used in hazardous applications like *explosion-proof apparatus* because of no electrical circuit.
- Moreover, there is no possibility of *spark production* as well.

Vacuum Grippers

- Vacuum grippers are also called as suction cups, that can be used for handling certain type of objects. Vacuum grippers can handle only flat, smooth, clean conditioned objects to form a satisfactory vacuum between the object and the suction cup.
- The vacuum gripper's suction cup is made of elastic material such as rubber or soft plastic. While designing some means of removing the air between the cup and the part surface and to create a vacuum, vacuum pump and the venturi are the common devices used.
- The *Vacuum pump* is piston operated and powered by an electric motor. It is capable of creating a relatively high vacuum. The *Venturi* is driven by “shop air pressure”. Its initial cost is less than that of a vacuum pump and it is relatively reliable because of its simplicity.

Vacuum Grippers (Cond.)



Hooks (Non- Mechanical)

- Hooks can be used as the end effector to handle containers and to load or unload parts hanging from overhead conveyors.
- The item or object to be handled by a hook must have some sort of handle, so that hook can handle it.



Special Tools



Arc Welding



Machining



Spray Painting

Spray Painting Tool

- Spray painting tools are used by robots to apply coatings, paints, or finishes to surfaces with precision and consistency.
- These end effectors typically consist of a paint gun or sprayer attached to the robot's arm, along with a supply of paint and compressed air.
- Robots equipped with spray painting tools are widely used in automotive manufacturing, aerospace, furniture production, and other industries requiring high-quality surface finishing.

Arc Welding

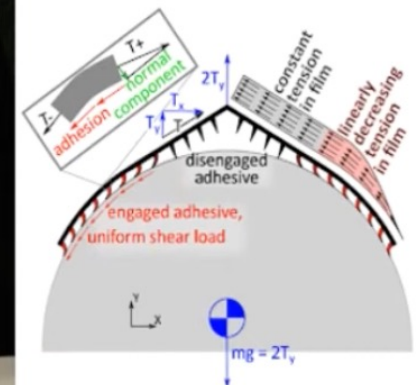
- Arc welding torches are tools used by robots to perform welding operations, including arc welding, spot welding, and seam welding.
- These end effectors typically include a welding torch or gun, a wire feeder, and a power supply.
- Robots equipped with arc welding torches are commonly used in automotive assembly, metal fabrication, shipbuilding, and construction, where precise and efficient welding is required.

Machining

- Machining tools, such as milling cutters, drills, grinders, and deburring tools, are used by robots to perform machining operations, including milling, drilling, grinding, and surface finishing.
- These end effectors are often equipped with cutting tools or abrasive wheels mounted on robotic arms.
- Robots equipped with machining tools are used in industries such as aerospace, automotive, metalworking, and precision engineering to automate manufacturing processes and improve efficiency and accuracy.

Adhesive Grippers

- An adhesion gripper is a robot end effector that grasps objects by literally sticking to them. In its most primitive form, this type of gripper consists of a rod, sphere, or other solid object covered with two-sided tape. Velcro can also be used if the object(s) to be grasped are likewise equipped.
- A major asset of the adhesive gripper is the fact that it is simple. As long as the adhesive keeps its "stickiness," it will continue to function without maintenance. However, there are certain limitations. The most significant is the fact that the adhesive cannot readily be disabled in order to release the grasp on an object. Some other means, such as devices that lock the gripped object into place, must be used.



Active Grippers

- Active grippers are equipped with actuators that actively control the opening and closing of the gripper fingers or jaws. These actuators provide the necessary force and motion to grasp and hold objects securely.
- Active grippers can be actuated using various mechanisms, including pneumatic, hydraulic, electric, or servo-driven systems. Each actuation method offers unique advantages in terms of speed, force, precision, and control.
- Advantages of active grippers include precise control over grip force and position, adaptability to different object shapes and sizes, and the ability to exert sufficient force to grasp heavy or irregularly shaped objects.
- Active grippers are commonly used in industrial automation, manufacturing, assembly lines, and robotic applications where precise manipulation and control are required.

Passive Grippers

- Passive grippers, also known as compliant grippers or non-actuated grippers, do not have built-in actuators for controlling finger movement. Instead, they rely on mechanical or material properties to achieve gripping and holding.
- Passive grippers may utilize compliant mechanisms, such as springs, flexures, or elastomers, to provide flexibility and adaptability in grasping objects with varying shapes and sizes. Passive grippers can also employ passive damping or friction mechanisms to enhance grip stability and prevent slippage during manipulation tasks.
- Advantages of passive grippers include simplicity, reliability, and cost-effectiveness. They are often lightweight, low-profile, and easy to integrate into robotic systems without complex control systems or power sources. Passive grippers are suitable for applications where delicate handling, gentle interaction with objects, or minimal power consumption is desired. They are commonly used in fields such as food handling, medical robotics, soft robotics, and collaborative robotics.

Active & Passive Grippers

- Active Gripper



Isaacc Asimov's 3 laws of Robotics

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey orders given it by human beings except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

Applications of Robots

- Robots are best suited for applications involving repetitive tasks, job requiring capabilities and precision beyond those of humans, and working in hazardous environments.
- Robots have already been used in many industries and for many purposes. They have excelled when they can perform better than humans or at lower costs.
- For example, a welding robot can probably weld better than a human welder in routine applications because it can move more uniformly and consistently along a prescribed path without needing protective goggles and clothing, ventilation, or other necessities.
- As a result, the robot can be more productive and better suited for the job as long as the task is set up for the robot, nothing happens to the setup, and the job is not too complicated. Similarly, a robot exploring the ocean floor requires far less attention than a human diver, can stay underwater for long periods of time, can go to very great depths and still survive the pressure, and yet does not require oxygen.

Robotic Applications

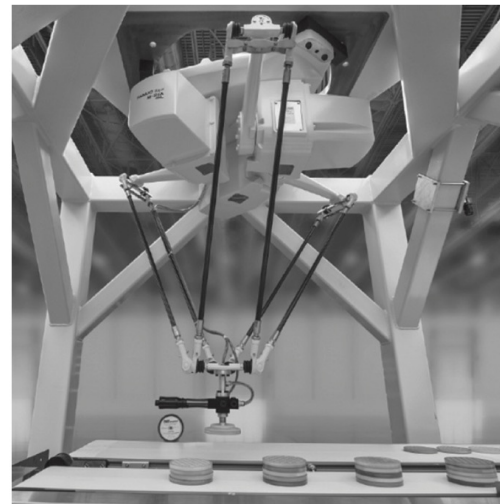
- **Machine loading:** Robots supply other machines with parts or remove the processed parts from other machines.
- In this type of work, the robot may not even perform any operation on the part, but rather facilitates material and parts handling and loading other machines within the context of a task.



Robots performing loading and unloading of parts.

Robotic Applications (Contd.)

- **Pick and place operations:** The robot picks up parts and places them elsewhere.
- This may include palletizing, placement of cartridges, simple assembly where two parts are put together (such as placing tablets into a bottle), placing parts in an oven and removing the treated parts from the oven, or other similar routines.



(a)

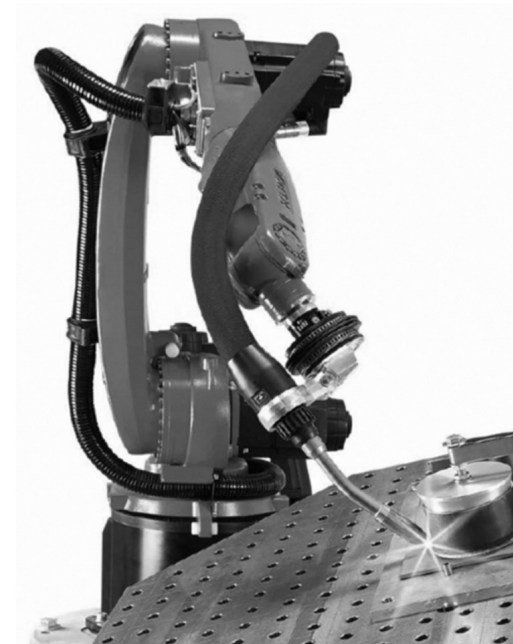


(b)

(a) A parallel robot stacking cookies. (b) A serial robot handling chocolate bars on a conveyor

Robotic Applications (Contd.)

- **Welding:** The robot, along with proper setups and a welding end effector, is used to weld parts together.
- This is one of the most common applications of robots in the auto industry.
- Due to their consistent movements, robotic welds are very uniform and accurate. Welding robots are usually large and powerful



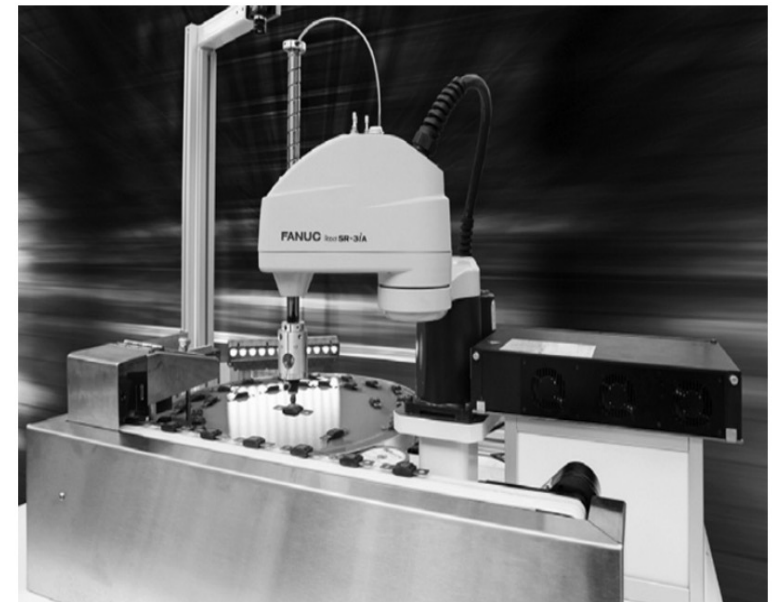
A robot welding parts together.

Robotic Applications (Contd.)

- **Painting:** Is another very common application of robots, especially in the automobile industry.
- Since maintaining a ventilated but clean paint area suitable for humans is difficult to achieve, and because compared to human's robotic operations are more consistent, painting robots are very well suited for their job.

Robotic Applications (Contd.)

- **Inspection:** Inspection of parts, circuit boards, and other similar products is also a very common application for robots. In general, robots are one component of an inspection system that may include a vision system, an X-ray device, an ultrasonic detector, or other similar devices.
- In one application, a robot was equipped with an ultrasonic crack detector, was given the CAD data about the shape of an airplane fuselage and wings and was used to follow the airplane's body contours and check each joint, weld, or rivet.



A robot engaged in sorting and inspection of manufactured parts.

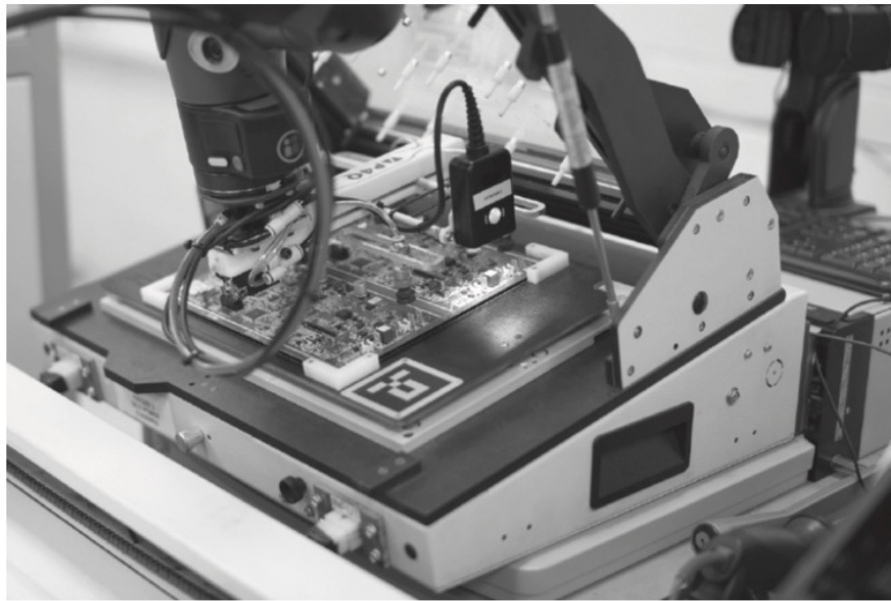
Robotic Applications (Contd.)

- In a similar application, a robot would search for and find the location of each rivet, detect and mark the rivets with fatigue cracks, drill them out, and move on.
- The technicians would insert and install new rivets. Robots have also been extensively used for circuit board and chip inspection. In most applications like this, including part identification, the characteristics of the part (such as the circuit diagram of a board, the nameplate of a part, and so on) are stored in the system in a data library.
- The system uses this information to match the part with the stored data. Based on the result of the inspection, the part is either accepted or rejected.
- **Sampling:** Sampling with robots is used in the agriculture industry, as well as in many other industries. Sampling can be like pick-and-place and inspection, except that it is performed only on a certain number of products.

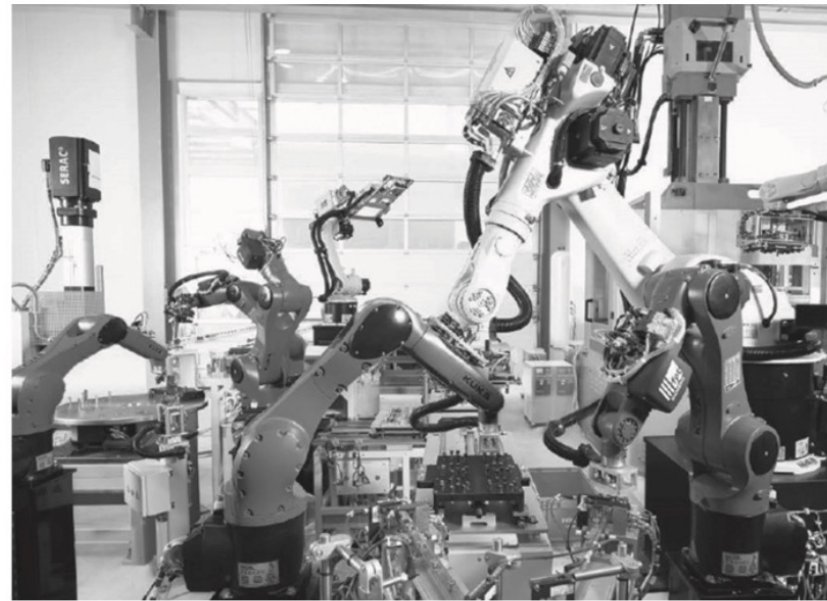
Robotic Applications (Contd.)

- **Assembly tasks:** Usually involve many operations. For example, the parts must be located and identified, they must be carried in a particular order with many obstacles around the setup, and they must be fit together and then assembled.
- Many of the fitting and assembling tasks are complicated and may require pushing, turning, bending, wiggling, pressing, snapping tabs to connect the parts, and other operations. Slight variations in parts and their dimensions due to larger tolerances also complicate the process, since the robot has to know the difference between variations in parts and wrong parts.

Robotic Applications (Contd.)



(a)



(b)

(a) A Sawyer robot inserting electronic parts into a circuit with its specialized end effector. (b) A group of KUKA robots on an assembly line.

Robotic Applications (Contd.)

- **Manufacturing** by robots may include many different operations, such as material removal, drilling, deburring, laying glue, cutting, and so on.
- It also includes insertion of parts, such as electronic components into circuit boards, installation of boards into electronic devices, and other similar operations.
- Insertion robots are very common and are extensively used in the electronic industry.



A robot engaged in a manufacturing task.

Robotic Applications (Contd.)

- Medical robotics refers to the use of robotic systems and technologies in various aspects of healthcare, ranging from surgery and rehabilitation to diagnostics and patient care. These applications aim to improve the quality, efficiency, and precision of medical procedures while also enhancing patient outcomes. Here are some common medical applications of robotics:
 - 1. Robotic Surgery:** Robotic surgical systems, such as the da Vinci Surgical System, enable surgeons to perform minimally invasive procedures with greater precision and dexterity. These systems use robotic arms controlled by surgeons to perform complex manoeuvres with smaller incisions, leading to reduced pain, shorter hospital stays, and faster recovery times for patients.
 - 2. Telepresence and Remote Surgery:** Teleoperated robotic systems allow surgeons to perform procedures on patients located at a different location. This is particularly useful for remote and underserved areas where access to specialized healthcare is limited. Telepresence robots also facilitate remote consultations, medical education, and training.

Robotic Applications (Contd.)

3. **Rehabilitation Robotics:** Rehabilitation robots assist patients in regaining mobility and functionality after injury or surgery. These robots can provide repetitive and targeted therapy exercises to improve muscle strength, coordination, and range of motion. Examples include robotic exoskeletons for walking assistance and robotic arms for upper limb rehabilitation.
4. **Assistive Robotics:** Assistive robots support individuals with disabilities or age-related impairments in performing daily activities and tasks. These robots can help with mobility, personal care, medication management, and household chores, thereby enhancing independence and quality of life for users.
5. **Diagnostic and Imaging Robotics:** Robotic systems are used in medical imaging and diagnostic procedures to improve accuracy and efficiency. For example, robotic MRI-guided biopsy systems enable precise targeting of tissue samples for analysis, reducing the need for invasive procedures and increasing diagnostic yield.

Robotic Applications (Contd.)

6. **Pharmacy and Drug Delivery:** Automated pharmacy systems and robotic drug dispensing systems help pharmacists and healthcare providers accurately prepare and dispense medications. These systems reduce errors, improve medication adherence, and enhance medication management processes in healthcare facilities.
7. **Surgical Training and Simulation:** Virtual reality (VR) and augmented reality (AR) technologies combined with robotic simulators provide realistic training environments for surgeons and medical professionals. These simulation platforms allow learners to practice surgical techniques, refine skills, and familiarize themselves with new procedures in a safe and controlled setting.
8. **Nanorobotics:** Nanorobots, or nanobots, are miniature robots designed to perform tasks at the nanoscale, such as targeted drug delivery, molecular imaging, and tissue engineering. While still in the experimental stage, nanorobotics holds promise for revolutionizing drug delivery and precision medicine in the future.

THE END
THANK YOU



Department Of Artificial Intelligence & Data Science